ELEC70013
EE9-AO4

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

HIGH PERFORMANCE ANALOGUE ELECTRONICS

Monday, 12 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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The Questions

1. A radio communication system uses a mixer to down-convert a high-frequency signal to an intermediate frequency (IF). The input signal has a frequency of 900 MHz, and the local oscillator (LO) signal has a frequency of 860 MHz.

(a) Explain the primary purpose of a mixer in a radio communication system.

[2]

(b) Describe the two main output frequencies produced by the mixing process and explain the mathematical basis behind them.

[3]

(c) Calculate the two output frequencies produced by the mixer when the input signal and the LO signal are combined. Identify which of these frequencies would typically be selected as the intermediate frequency (IF) and explain why.

[5]

(d) Define the concept of an image frequency in the context of a mixer.

[2]

(e) If the IF is chosen to be 40 MHz, calculate the image frequency and explain why it might need to be filtered out.

[3]

(f) Explain the role of a mixer in a superheterodyne receiver. Discuss how the choice of intermediate frequency (IF) affects the receiver's performance in terms of image rejection.

[5]

2.

(a) State the transfer functions for biquad filters.

[2]

(b) Explain the significance of having a linear phase response in filter design. Why could this be important in audio processing?

[2]

(c) The schematic below corresponds to a fifth order low pass filter. Draw its equivalent circuit implemented with gyrators based on transconductors.

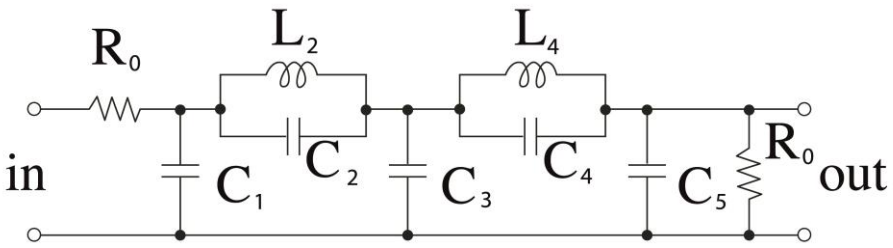


Figure 2.1

[3]

(d) Given the following state-space representation of an LC ladder filter:

$$\dot{x}(t) = A\,x(t) + B\,u(t)$$
$$y(t) = C\,x(t) + D\,u(t)$$

where:

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = [\,1 \;\; 0\,], \quad D = [\,0\,]$$

Derive the transfer function of the system from the given state-space equations. Determine whether the filter behaves as a low-pass, high-pass, band-pass, or band-stop filter. Explain your reasoning.

[3]

(e) The equation below represents a second order Butterworth filter, designed with inductors and capacitors. What should be the value of L an C for a cut-off frequency of 1MHz? Reminder: Cut-off frequency represents the voltage drop to 70.7% of the passband value.

$$H(s) = \frac{1}{LCs^2 + \left(\frac{1}{C}\right)s + 1}$$

[3]

(f) If the filter above was to be implemented with transconductor based gyrators, and the available transconductance is 100 µS what value capacitors would be needed for the gyrator?

[3]

(g) If the capacitor values were fixed instead, explain how the transconductance value influences the cutoff frequency and what adjustments you would make if the cutoff frequency were slightly higher than the target value.

[4]

3.

(a) Derive the output current of a MOS differential pair in weak inversion.

[2]

(b) Explain descriptively, but with a mathematical argument, why differential circuits are better in terms of linearity than single ended ones.

[2]

(c) The transfer function of a differential transconductor is:

$$I_{out} = a\, V_{CM} + \beta\, V_d\, e^{(cV_{G1})} - dV_T;$$

where V_{CM} represents the input common mode voltage, V_d the differential voltage, and V_{G1} the constant voltage in the gate of a transistor operating in weak inversion, while V_T is a technological parameter.

Describe which terms are non-ideal from the perspective of transconductor performance and explain the reasons why they affect performance.

[4]

(d) Describe how would you modify the transconductor above to improve its performance. Give an expression for the output current of whatever you propose.

[4]

(e) Is there any technique that could be considered to make a weak inversion transconductor fully linear in large signal. Explain your reasoning.

[4]

(f) How would you improve the transconductor below to increase its linearity? Give an expression for the output current of whatever you propose.

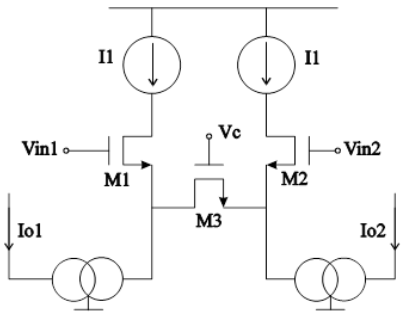


Figure 3.1

[4]

4.

(a) For the circuit below, if M1 is operating in the strong inversion ohmic region and the amplifier is ideal, give an expression for I_{out}

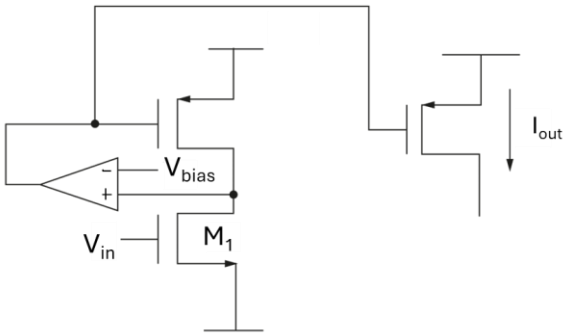


Figure 4.1

[3]

(b) Give an expression for the large signal transconductance for the circuit above.

[3]

(c) For the circuit in Figure 4.2 give an expression for I_{out} assuming the operating region and sizes for the transistors that would optimize linearity.

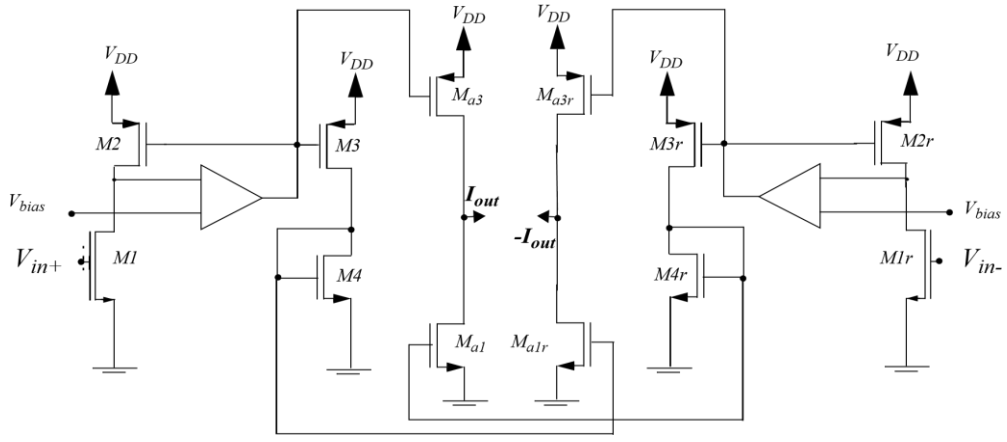


Figure 4.2

[4]

(d) What is the value of the transconductance for the circuit in Figure 4.2.

[3]

(e) How would you modify the circuit in Figure 4.2 to be an integrator?

[3]

(f) For the circuit in the figure below give an expression for I_{out} explaining your assumptions and rationale.

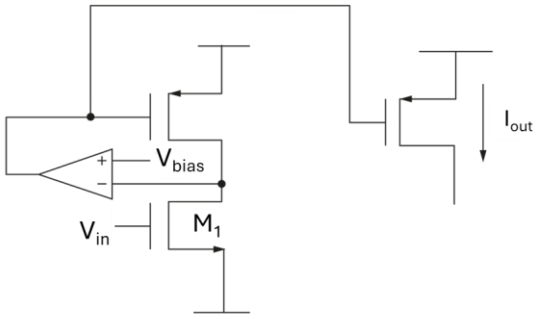


Figure 4.3

[4]